REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Practice Questions and Worked Answers

Grade 9 Advanced | Module 2 Lesson 4 | How Fast

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: $v_{avg} = \Delta x/\Delta t$; average speed = total distance/total time; $\Delta x = v\Delta t$ for constant velocity.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 9 book, printed pages 42–48.

P01. Average velocity is calculated using:

A. Displacement divided by time interval B. Distance multiplied by time
C. Speed divided by distance D. Final position alone

Answer

A. Displacement divided by time interval.

P02. Which quantity is always nonnegative?

A. One-dimensional velocity B. Speed
C. Displacement D. Position coordinate

Answer

B. Speed.

P03. Instantaneous velocity is represented on a position–time graph by:

A. Total graph height divided by mass B. Area under the graph
C. Slope of the tangent at an instant D. The time-axis intercept alone

Answer

C. Slope of the tangent at an instant.

P04. A round trip can have:

A. Negative total distance B. Nonzero displacement with identical endpoints
C. Zero travel time D. Zero average velocity but nonzero average speed

Answer

D. Zero average velocity but nonzero average speed.

P05. Uniform velocity means constant:

A. Speed and direction B. Speed only with any direction
C. Position D. Acceleration of any value

Answer

A. Speed and direction.

P06. An object moves from -6 m to 12 m in 8 s. Calculate its average velocity.

A. 2.25 m/s B. 4.5 m/s
C. 1.125 m/s D. -2.25 m/s

Answer

A. 2.25 m/s. vavg = Δx/Δt = (12 − (-6))/8 = 2.25 m/s.

P07. A runner covers 12 m outward and 8 m back in 10 s. Calculate the average speed.

A. 4 m/s B. 2 m/s
C. 1 m/s D. -2 m/s

Answer

B. 2 m/s. Average speed = total distance/time = (12 + 8)/10 = 2 m/s.

P08. An object moves at constant velocity +6 m/s for 6 s. Calculate its displacement.

A. 72 m B. 18 m
C. 36 m D. -36 m

Answer

C. 36 m. Δx = vΔt = 6 × 6 = +36 m.

P09. A cyclist travels 120 m at constant speed 10 m/s. Calculate the travel time.

A. 24 s B. 6 s
C. -12 s D. 12 s

Answer

D. 12 s. t = d/v = 120/10 = 12 s.

P10. Convert a speed of 72 km/h to m/s.

A. 20 m/s B. 40 m/s
C. 10 m/s D. -20 m/s

Answer

A. 20 m/s. v = 72 × 1000/3600 = 20 m/s.

P11. Distinguish average speed from average velocity.

Answer

Average speed uses total distance/time; average velocity uses displacement/time and includes direction.

P12. Define instantaneous velocity.

Answer

Velocity at a particular instant, found as the tangent slope of a position–time graph.

P13. When does average speed equal the magnitude of average velocity?

Answer

When the object does not reverse direction during the interval, so distance equals displacement magnitude.

P14. What does a negative average velocity mean in one dimension?

Answer

The net displacement is in the chosen negative direction during a positive time interval.

P15. Why must total time include stops when calculating overall average speed?

Answer

The entire trip duration includes stopped intervals, which affect the total distance divided by total time.

P16. An object moves from -9 m to 18 m in 12 s. Calculate its average velocity.

Answer

$v_{avg} = \Delta x/\Delta t = (18 - (-9))/12 = 2.25$ m/s.

P17. An object moves from -12 m to 24 m in 16 s. Calculate its average velocity.

Answer

$v_{avg} = \Delta x/\Delta t = (24 - (-12))/16 = 2.25$ m/s.

P18. A runner covers 18 m outward and 12 m back in 15 s. Calculate the average speed.

Answer

Average speed = total distance/time = (18 + 12)/15 = 2 m/s.

P19. A runner covers 24 m outward and 16 m back in 20 s. Calculate the average speed.

Answer

Average speed = total distance/time = (24 + 16)/20 = 2 m/s.

P20. An object moves at constant velocity +9 m/s for 7 s. Calculate its displacement.

Answer

$\Delta x = v\Delta t = 9 \times 7 = +63$ m.

P21. An object moves at constant velocity +12 m/s for 8 s. Calculate its displacement.

Answer

$\Delta x = v\Delta t = 12 \times 8 = +96$ m.

P22. A cyclist travels 180 m at constant speed 15 m/s. Calculate the travel time.

Answer

$t = d/v = 180/15 = 12$ s.

P23. A cyclist travels 240 m at constant speed 20 m/s. Calculate the travel time.

Answer

$t = d/v = 240/20 = 12$ s.

P24. Convert a speed of 108 km/h to m/s.

Answer

$v = 108 \times 1000/3600 = 30$ m/s.

P25. Convert a speed of 144 km/h to m/s.

Answer

$v = 144 \times 1000/3600 = 40$ m/s.

P26. A runner travels 100 m east in 20 s and returns 100 m west in 30 s. Find overall average speed and average velocity.

Answer

Total distance = 200 m and total time = 50 s, so average speed = 4.0 m/s. Displacement = 0, so average velocity = 0.

P27. A student averages two speeds, 4 m/s and 8 m/s, for two equal-distance legs. Is 6 m/s the trip average? Explain.

Answer

No. Equal distances take unequal times. For distance d on each leg, vavg = 2d/(d/4 + d/8) = 16/3 ≈ 5.33 m/s.

P28. A position–time curve is tangent to a horizontal line at one instant. What is the instantaneous velocity then?

Answer

The tangent slope is zero, so instantaneous velocity is zero. This alone does not show that the object remains at rest afterward.

P29. Explain why a speedometer reading does not necessarily equal the average speed of a long trip.

Answer

It measures speed near the current instant. The trip average depends on all distances and elapsed times, including stops and speed changes.

P30. Design a simple method to measure a student's average walking speed along a straight corridor.

Answer

Measure a known distance, time the whole walk, calculate distance/time and repeat. State start and finish points clearly and control timing consistency.